Beyond the Prompt

Building a geospatial research project with Codex — from a broad question to a reproducible, bounded experiment.

THE QUESTION

Which summits offer the most interesting mountain panoramas — and does height predict the answer?

A human supplied the objective, local knowledge, source files and scope decisions. Codex inspected the environment, wrote and ran Python, tested the methods and generated the deliverables. Development used code; no manual QGIS or Figma export was required.

1,302

Existing observation summits; Days 01–04 preserved.

2 + 1

Two reports and one static interactive explorer.

Human responsibilities: set the scope, supply program identities, accept exploratory accuracy and authorize resource use. Agent responsibilities: implement, measure, preserve evidence and disclose uncertainty. Publication is a separate authorization.

PROMPT → EVIDENCE → PUBLICATION

01 Discover

Audit Windows, source data, memory and GIS runtimes.

02 Test

Compare engines; measure a pilot before full execution.

03 Compute

Checkpoint every summit; control concurrency live.

04 Interpret

Freeze component formulas, then rank and test sensitivity.

05 Export

Python builds PDF, PNG, JSON, CSV and static web assets.

How the code became a pipeline

The important revisions changed both the scientific scope and the cost of execution.

G1 · Engine

GDAL synthetic acceptance; correct CRS, heights, curvature and NoData.

G2 · Inputs

Complete 100 km DEM support; international peak targets; source hashes.

G3 · Benchmark

Twelve sequential cases; measure time and memory rather than guess.

G4 · Production

1,302 checkpoints; three workers; pause or reduce concurrency at any time.

G5 · Rankings

Frozen NPS; scoped Horizon-only MPS; weight and skyline sensitivity.

G6 · Exports

Reproducible reports; browser-only ranking weights; local staging.

Actual prompt revisions

“considering dramatic speedup and that this is not real research project but more for fun right now, let's re plan the pipeline using GDAL”

“retain only 1 GiB reserved, test first with 3 workers, then after successfull test start G4 full analysis. While processing G4, make sure to have checkpoints in order to reduce number of workers anytime”

The supplied prompt archive records these revisions. Source hashes, per-summit artifacts and machine-readable CLI logs connect a decision to the result it produced. Counts below refer only to captured records.

What worked. What failed. What remains.

Measured evidence is reported separately from missing development telemetry.

2828
Recorded process starts, including workers

16
Supplied prompt files captured

4.61 h
G4 elapsed wall time, including an interruption gap

1.56 GiB
Sampled peak G4 process-tree RSS

Reliable patterns

Small pilots prevented expensive wrong turns. Hashed checkpoints let work resume safely. The six-case three-worker benchmark measured a 1.55× speedup; concurrency could drain from three workers to one and return without losing completed results.

Useful failures

90 m terrain changed visible area by roughly 20–32% in the pilot, so 30 m was retained. Windows installation needed a short-prefix prebuilt environment and a GRASS DLL workaround. GDAL was accepted with disclosed disagreement. RAM gates paused later attempts; no unrelated processes were killed.

Honest boundaries

Complete agent-iteration counts and active development hours were not captured. CLI starts are not agent turns. Log durations overlap and must not be summed as elapsed development. GIS tests are not field validation. Rankings depend on scope and subjective weights.

Reproducibility is a deliverable

At export: 2771 successful and 50 failed captured CLI finishes. Earlier baseline verification: 51 tests passed; MPS: 3 focused tests passed. G6 browser/export validation is saved separately with the build. Source locks, manifests, actual prompts, decisions and failure records are retained.

Lesson: a prompt-only workflow still needs explicit human choices, scientific restraint and reproducible checks. The useful output is an inspectable chain from source data to a result, with its limits attached.